YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 4 SOLUTIONS

Trigonometry (CAS)

1 a The unit circle has a radius of one.

The line representing $\sin(\theta)$ is parallel to the y-axis and the line representing $\cos(\theta)$ is parallel to the x-axis so these two lines are perpendicular.

Use Pythagoras' theorem.

$\sin^2(\theta) + \cos^2(\theta) = 1^2 = 1$ **[2]**

b $\sin^2(\theta) + \cos^2(\theta) = 1$

$\cos^2(\theta) = 1 - \sin^2(\theta) = [1 - \sin(\theta)][1 + \sin(\theta)]$

$\cos^2(\theta) = [1 - \sin(\theta)][1 + \sin(\theta)]$ **[2]**

c $\cos^2(\theta) = [1 - \sin(\theta)][1 + \sin(\theta)]$

Divide both sides by $\cos(\theta)[1 - \sin(\theta)]$.

$$\frac{\cos^2(\theta)}{\cos(\theta)[1 - \sin(\theta)]} = \frac{[1 - \sin(\theta)][1 + \sin(\theta)]}{\cos(\theta)[1 - \sin(\theta)]}$$

$$\frac{\cos(\theta)}{[1 - \sin(\theta)]} = \frac{[1 + \sin(\theta)]}{\cos(\theta)}$$ **[2]**

This is not valid if $\cos(x) = 0$ or $\sin(x) = 1$, i.e. $x = \left(\frac{\pi}{2}\right) + n\pi$ **[1]**

The identity is valid if $x \neq \frac{\pi}{2} + n\pi$ **[1]**

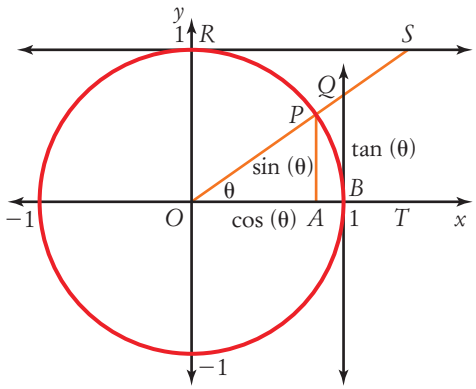
d Radius $= OR = OP = OB = 1$

In similar triangles OQB and OPA

$$\frac{OQ}{OB} = \frac{OP}{OA}$$

$$\frac{OQ}{1} = \frac{1}{\cos(\theta)}$$

so $OQ = \sec(\theta)$



[2]

In similar triangles OSR and POA

$$\frac{RS}{BO} = \frac{OR}{QB}$$
$$\frac{RS}{1} = \frac{1}{\tan(\theta)}$$
$$\text{so } RS = \cot(\theta)$$

[2]

$$\frac{OS}{OR} = \frac{PO}{PA}$$
$$\frac{OS}{1} = \frac{1}{\sin(\theta)}$$
$$\text{so } OS = \operatorname{cosec}(\theta)$$

[2]

The second result shows that the cotangent is the tangent of the complementary angle.

2 a $\sec(45°) = \dfrac{1}{\cos(45°)} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$ **[2]**

b $\cot\left(\dfrac{\pi}{6}\right) = \dfrac{1}{\tan\left(\dfrac{\pi}{6}\right)} = \dfrac{3}{\sqrt{3}} = \sqrt{3}$ **[2]**

c Expand $\sin\left(\dfrac{\pi}{3} - \dfrac{\pi}{4}\right)$ or $\sin\left(\dfrac{\pi}{4} - \dfrac{\pi}{6}\right)$ **[1]**

$$\sin\left(\frac{\pi}{12}\right) = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$$
$$= \sin\left(\frac{\pi}{3}\right)\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{3}\right)$$
$$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \times \frac{1}{2}$$
$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

[2]

OR

$$\sin\left(\frac{\pi}{12}\right) = \sin\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$$
$$= \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{6}\right) - \sin\left(\frac{\pi}{6}\right)\cos\left(\frac{\pi}{4}\right)$$
$$= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{\sqrt{2}}{2}$$
$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

[2]

d $\dfrac{2\tan\left(\dfrac{3\pi}{8}\right)}{1 - \tan^2\left(\dfrac{3\pi}{8}\right)} = \tan\left(2 \times \dfrac{3\pi}{8}\right) = \tan\left(\dfrac{3\pi}{4}\right) = -1$ **[2]**

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e Let $x = \tan\left(\frac{3\pi}{8}\right)$.

$$\frac{2x}{1-x^2} = -1$$
$$2x = -1 + x^2$$
$$x^2 - 2x + 1 = 0$$ **[2]**

Complete the square. OR Use the quadratic formula.

$x^2 - 2x + 1 = 1 + 1$
$(x-1)^2 = 2$

$a = 1, b = -2, c = -1$
$x = \frac{2 \pm \sqrt{4+4}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}$

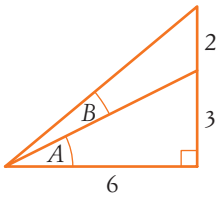
$x = 1 \pm \sqrt{2}$ **[2]**

Since $\frac{3\pi}{8} < \frac{\pi}{2}$, $\frac{3\pi}{8}$ is in the first quadrant, so $\tan\left(\frac{3\pi}{8}\right) > 0$ **[1]**

Hence $\tan\left(\frac{3\pi}{8}\right) = 1 + \sqrt{2}$ **[1]**

f Let $C = A + B$ (optional)

$\tan(C) = \frac{5}{6}$ and $\tan(A) = \frac{3}{6} = \frac{1}{2}$ **[1]**



$\tan(B) = \tan(C - A)$
$= \frac{\tan(C) - \tan(A)}{1 + \tan(C)\tan(A)}$
$= \frac{\frac{5}{6} - \frac{3}{6}}{1 + \frac{5}{6} \times \frac{1}{2}}$
$= \frac{\frac{2}{6}}{1 + \frac{5}{12}} \times \frac{12}{12}$
$= \frac{4}{17}$

OR

$\tan(B) = \tan[(A + B) - A]$
$= \frac{\tan(A+B) - \tan(A)}{1 + \tan(A+B)\tan(A)}$
$= \frac{\frac{5}{6} - \frac{3}{6}}{1 + \frac{5}{6} \times \frac{1}{2}}$
$= \frac{\frac{2}{6}}{1 + \frac{5}{12}} \times \frac{12}{12}$
$= \frac{4}{17}$

[2]

g $\sin(75°) - \sin(15°) = 2\sin\left(\frac{90°}{2}\right)\cos\left(\frac{60°}{2}\right)$

$= 2\sin(30°)\cos(45°)$

$= 2 \times \frac{1}{2} \times \frac{\sqrt{2}}{2}$

$= \frac{\sqrt{2}}{2}$ **[2]**

h $\cos(85°) + \cos(155°) + \cos(35°) = 2\cos(120°)\cos(35°) + \cos(35°)$

$= -\cos(35°) + \cos(35°)$

$= 0$ **[2]**

i $2\sin(15°)\sin(75°) = \cos(60°) - \cos(90°) = \frac{1}{2}$

$\sin(15°)\sin(75°) = \frac{1}{4}$ **[2]**

3 a $\frac{\cot(x)}{1+\cot(x)} = \frac{\frac{1}{\tan(x)}}{1+\frac{1}{\tan(x)}}$ **[1]**

$= \frac{\frac{1}{\tan(x)}}{\frac{\tan(x+1)}{\tan(x)}}$

$= \frac{1}{\tan(x)} \times \frac{\tan(x)}{\tan(x+1)}$

$= \frac{1}{\tan(x+1)}$ **[1]**

OR

$\frac{\cot(x)}{1+\cot(x)} = \frac{\cot(x)}{1+\cot(x)} \times \frac{\tan(x)}{\tan(x)}$ **[1]**

$= \frac{1}{\tan(x+1)}$ **[1]**

b $\dfrac{1+\sin(P)-\cos^2(P)}{\cos(P)[1+\sin(P)]} = \dfrac{1-\cos^2(P)+\sin(P)}{\cos(P)[1+\sin(P)]}$

$= \dfrac{\sin^2(P)+\sin(P)}{\cos(P)[1+\sin(P)]}$ **[1]**

$= \dfrac{\sin(P)[\sin(P)+1]}{\cos(P)[1+\sin(P)]}$ **[1]**

$= \dfrac{\sin(P)}{\cos(P)}$ **[1]**

$= \tan(P)$ **[1]**

[4 in total]

c Simplify $\sin(\theta)+\sin(3\theta)$ and $\cos(\theta)+\cos(3\theta)$. **[1]**

$\dfrac{\sin(\theta)+\sin(2\theta)+\sin(3\theta)}{\cos(\theta)+\cos(2\theta)+\cos(3\theta)} = \dfrac{2\sin(2\theta)\cos(\theta)+\sin(2\theta)}{2\cos(2\theta)\cos(\theta)+\cos(2\theta)}$ **[1]**

$= \dfrac{\sin(2\theta)[2\cos(\theta)+1]}{\cos(2\theta)[2\cos(\theta)+1]}$ **[1]**

$= \dfrac{\sin(2\theta)}{\cos(2\theta)}$ **[1]**

$= \tan(2\theta)$ **[1]**

[4 in total]

d $\tan(2y)+\sec(2y) = \dfrac{\sin(2y)}{\cos(2y)} + \dfrac{1}{\cos(2y)}$

$= \dfrac{\sin(2y)+1}{\cos(2y)}$ **[2]**

$= \dfrac{2\sin(y)\cos(y)+\sin^2(y)+\cos^2(y)}{\cos^2(y)-\sin^2(y)}$ **[2]**

$= \dfrac{\left[\sin(y)+\cos(y)\right]^2}{\left[\cos(y)-\sin(y)\right]\cos(y)-\sin(y)}$ **[1]**

$= \dfrac{\sin(y)+\cos(y)}{\cos(y)-\sin(y)}$

$\tan(2y)+\sec(2y) = \dfrac{\cos(y)+\sin(y)}{\cos(y)-\sin(y)}$ **[1]**

[5 in total]

4 a $\sin(4\theta) = 2\sin(2\theta)\cos(2\theta)$

$= 4\sin(\theta)\cos(\theta)[1-2\sin^2(\theta)]$ **[2]**

Since $\cos(\theta) = \pm\sqrt{1-\sin^2(\theta)}$ then this is not possible. **[1]**

b Use $\sin(4\theta) = 4\sin(\theta)\cos(\theta) - 8\sin^3(\theta)\cos(\theta)$ (from part **a**)

$$\begin{aligned}\sin(5\theta) = \sin(4\theta + \theta) &= \sin(4\theta)\cos(\theta) + \cos(4\theta)\sin(\theta)\\ &= [4\sin(\theta)\cos(\theta) - 8\sin^3(\theta)\cos(\theta)]\cos(\theta) + [1 - 2\sin^2(2\theta)]\sin(\theta)\\ &= 4\sin(\theta)\cos^2(\theta) - 8\sin^3(\theta)\cos^2(\theta) + \sin(\theta) - 2\sin^2(2\theta)\sin(\theta)\\ &= 4\sin(\theta)[1 - \sin^2(\theta) - 8\sin^3(\theta)][1 - \sin^2(\theta)] + \sin(\theta) - 2[2\sin(\theta)\cos(\theta)^2\sin(\theta)]\\ &= 4\sin(\theta) - 4\sin^3(\theta) - 8\sin^3(\theta) + 8\sin^5(\theta) + \sin(\theta) - 8\sin^3(\theta)\cos^2(\theta)\\ &= 5\sin(\theta) - 12\sin^3(\theta) + 8\sin^5(\theta) - 8\sin^3(\theta)[1 - \sin^2(\theta)]\\ &= 5\sin(\theta) - 12\sin^3(\theta) + 8\sin^5(\theta) - 8\sin^3(\theta) + 8\sin^5(\theta)\\ &= 5\sin(\theta) - 20\sin^3(\theta) + 16\sin^5(\theta)\end{aligned}$$ **[5]**

Note: Students may start with $\sin(3\theta + 2\theta)$ or other possibilities.

c i $\sin\left(\frac{5\pi}{10}\right). = 5x - 20x^3 + 16x^5$ and $\sin\left(\frac{5\pi}{10}\right) = \sin\left(\frac{\pi}{2}\right) = 1.$ **[2]**

$16x^5 - 20x^3 + 5x = 1$ **[1]**

ii $16x^5 - 20x^3 + 5x - 1 = 0$ **[1]**

iii $(x - 1)(4x^2 + 2x - 1)^2 = 0$ **[1]**

$x = 1$ or $4x^2 + 2x - 1 = 0$

$\sin\left(\frac{\pi}{10}\right) \neq 1$ so eliminate $x = 1$. **[1]**

Solve $4x^2 + 2x - 1 = 0$

Use the quadratic formula.

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 4 \times (-1)}}{2 \times 4} = \frac{-2 \pm \sqrt{20}}{8} = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$ **[2]**

Since $\sin\left(\frac{\pi}{10}\right) > 0$, $\sin\left(\frac{\pi}{10}\right) = \frac{-1 + \sqrt{5}}{4}$ or $\frac{\sqrt{5} - 1}{4}$. **[1]**

d Given it works for $n = 1$ [$\sin(1\theta) = \sin(\theta)$], and $n = 5$ (part **b**), and it doesn't work for $n = 2$ [$\sin(2\theta) = 2\sin\cos(\theta)$], and or $n = 4$ (part **a**) you would expect it would work for odd values of n. Students may also recall formulae for $\sin(3\theta)$ and $\cos(3\theta)$, for which it is true. **[1]**

e The conjecture is that $\sin[(2n - 1)\theta] = f[\sin(\theta)]$ and $\cos[(2n - 1)\theta] = f[\cos(\theta)]$ where both are polynomials in odd powers of $\sin(\theta)$ and $\cos(\theta)$ respectively.

Test for $n = 1$: $\sin(1\theta) = \sin(\theta)$ which is a function of $\sin(\theta)$, with only an odd power (1).

$\cos(1\theta) = \cos(\theta)$ which is a function of $\cos(\theta)$, with only an odd power (1).

The conjecture is true for $n = 1$.

Assume true for $n = k$, i.e. $\sin[(2k - 1)\theta] = f[\sin(\theta)]$ and $\cos[(2k - 1)\theta] = g[\cos(\theta)]$.

Consider $n = k + 1$.

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$$\sin\{[2(k-1)+1]\theta\} = \sin[(2k-1)\theta]$$
$$= \sin[(2k-1)\theta + 2\theta]$$
$$= \sin[(2k-1)\theta + 2\theta] + \cos[(2k-1)\theta]\sin(2\theta)$$
$$= f[\sin(\theta)][1 - 2\sin^2(2\theta)] + g[\cos(\theta)]\,2\sin(\theta)\cos(\theta)$$

$f[\sin(\theta)][1 - 2\sin^2(\theta)]$ consists of odd powers of $\sin(\theta)$ multiplied by 1 and $\sin^2(\theta)$ which can only produce odd powers of $\sin(\theta)$.

$g[\cos(\theta)] \times \cos(\theta)$ consists of odd powers of $\cos(\theta)$ multiplied by $\cos(\theta)$, producing even powers of $\cos(\theta)$. These can be converted to even powers of $\sin(\theta)$ using $\cos^2(\theta) = 1 - \sin^2(\theta)$. When multiplied by $\sin(\theta)$ these even powers will all become odd powers.

Thus this part of the conjecture works for $n = k + 1$. **[4]**

The argument is basically the same for cosine.

$$\cos\{[2(k+1)-1]\theta\} = \cos[(2k+1)\theta]$$
$$= \cos[(2k-1)\theta + 2\theta]$$
$$= \cos[(2k-1)\theta]\cos(2\theta) + \sin[(2k-1)\theta]\sin(2\theta)$$
$$= g[\cos(\theta)][2\cos^2(\theta) - 1)] + f[\sin(\theta)]\,2\sin(\theta)\cos(\theta)$$ **[1]**

The first term consists of odd powers of $\cos(\theta)$ multiplied by 1 and $\cos^2(\theta)$ which can only produce odd powers.

The second product consists of odd powers of $\sin(\theta)$ multiplied first by $\sin(\theta)$, producing even powers of $\sin(\theta)$, which can be converted to even powers of $\cos(\theta)$ using $\cos^2(\theta) = 1 - \sin^2(\theta)$.When multiplied by $\cos(\theta)$ these even powers will all become odd powers. **[1]***

$\therefore$ if the conjecture is valid for $n = k$, it is valid for $n = k + 1$.

But, it is valid for $n = 0$ (so valid for $n = 1$, $n = 2$, etc.)

$\therefore$ it is valid for all natural numbers. **[1]**

* The reasoning is the same. If they get the reasoning for $\cos(\theta)$ but not $\sin(\theta)$ they should still be awarded 4 marks.

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